Data taking and analysis

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1 Dark rate data taking

1.1 Introduction

PMTs are amazing, they allow us to measure down to single photons, are reliable and easy to use. This comes at the price or being very fragile (and expensive) photo detectors that need to be handled with great care. You won't need to move them during this project but if you feel at any point that you need, please discuss it with us beforehand. These are very sensitive detectors, therefore, please avoid touching their shiny surface.

PMTs function through the process of electron avalanche, they are operated at very high negative voltage to make this process possible. However, large voltage can be dangerous, if mishandled, for both humans and equipment. With regards to PMTs, please be careful to properly close the dark box before operating them as any light more intense than a few hundred photons would surely break them. Additionally, please warm them up slowly, in increments of 500V max, waiting a few minutes between each step.

The procedure for taking data with the PMTs is explained in the helper sheet by the door. Please follow it carefully.

1.2 Set the Oscilloscope

The oscilloscope is doing the data acquisition. After you have switched it on, select 100u/s per division and 20mV/div. For now we are only using one PMT, the one connected to the fourth channel of the PMT. You can check the sampling rate of the scope by pressing acquire under the waveform label. It should be 4GSa/s when taking data but see how it changes as you increase the time per division and add/remove channels on the scope (by clicking on the corresponding number). Also check that you have the acquisition more as Normal in the Acquire menu. You can adjust the trigger level with the Trigger level knob.

For now, we are triggering the data acquisition (i.e. looking at one window) on a PMT pulse. Eventually the trigger will be provided by the Laser. To set-up the trigger, push the Edge button under Trigger. You can then select which channel to use as source and whether you want the trigger to be on the rising or lowering edge of the pulse.

Before taking data, I would recommend zooming in on the pulse to see them clearly overlap, identify cosmic muons and verify that all is in order. Then bring back the window to having a total of 1ms.

1.3 Launch the data collecting code

The data acquisition from the scope is controlled by the computer. The code to aquire 1 scope window of data lies in the `/home/a/usbtmc2`.

In any terminal use the `cd` command to change to the correct directory. The code to collect the data is `measure_spectra_1s_1channel.sh`. When you are in the correct folder you can read and modify any text-based file by doing the command `kwrite name_of_the_file`. Remember to save with CTRL+S before

you try running any of your modifications. You can keep a file open for modifications by calling CTRL+z in the terminal and then typing `bg` which stops the current process and lets it be active in the background. Alternatively you can append `&` at the and of any command to have it run in the background.

The `measure_spectra_1s_1channel.sh` code takes a total number of `c` windows of the scope. It was named 1s because we used to take 1000 windows of 1ms but one should start with smaller datasets to get comfortable with the procedure. It is based on the `get_mso_1.sh` script that communicates with the code and does not need to be modified. It first stores the scope settings as a .txt file, on line 13, this is only important for book keeping. It also moves the .dat file holding the raw data to the desired location and renames it so it doesn't get overwritten. Careful, if you do not use the 4th scope input you should change `chan4` to the one you are using.

The `bin2txt_v` executable that is compiled at the very beginning of the code is taking the data and the scope parameters (y_i) to calculate the correct data (converting the scope output to sensible values, depending on the number of V, s that were set per division on the scope). Careful, the `bin2txt_1channel.c` code that is compiled assumes a scope sampling rate of 4GHz for a 1ms total window duration. If you are working with different settings you will need to update the `BUFFERSIZE` parameter accordingly, where `BUFFERSIZE` is the total number of points expected for one scope window. The `bin2txt_1channel.c` code also stores and orders all the relevant data in a root file named `/home/a/WCTE540/PMT_data/out.root` that we then move (`mv`) where we want it and label it accordingly.

I couldn't stress enough how important a good file naming system is. A good file name should be easy to understand, not longer than it has to be and easy to call by subsequent codes. Please store your data in the `/home/a/WCTE540/Bsci_Project_T1_2023/Data` folder.

2 Preliminary data analysis

2.1 Combine the datasets

Once we have saved each of the window as a `.root` file we need to combine them to have significant statistics. This is done with the `hadd` command which is used (in the folder where you have the files that you want to combine) as follows: `hadd name_of_combined_file.root file_to_combine1 file_to_combine2 file_to_combine3 ....` A useful tool in Linux is the `*` symbol that allows you to call any file that contains the string that you put between asterix.

You will very likely run into the issue that the memory space in the machine in the lab is very limited. Once you have hadded all of your .root files into one I would make a copy of it (if it is an important file) on the external hard drive and delete all of the single-window root files. You can delete files in Linux with `rm name_of_file(s)`. You can list the files in the current directory with `ls`.

2.2 Check a dataset in the ROOT TBrowser

It is often useful to have a quick look at the data you have just taken to verify that nothing went wrong (e.g. PMT not plugged in, wrong data channel acquired, buffersize too small...). To do so, we use a `ROOT` interactive tool called a TBrowser that allows us to browse the content of a .root file and make histograms of the values stored in there.

To do so, first enter root by calling `root name_of_file_of_interest`. This opens a root instance in the terminal and attaches the file to it. you can then call `new TBrowser` to open the TBrowser window.

In the window, then click on your file that should be at the top of it. The information is held into a TTree called PMT1data holding branches which have the data. The histogram of the volts (i.e. how many points had what voltage) is the most useful to look at. You can change the y axis to being logarithmically-scaled

by right-clicking on the white bar on the left handside of the plot and clicking on the SetLogy option.

2.3 Perform the first clean up of the data

The data that was taken so far corresponds to all of the points that form the PMT waveform. most of these points aren't useful as they do not correspond to pulses but rather the stable background that we call the pedestal of the PMT.

This is too much useless data and blurs the 1pe signal (which is the charge collected by the PMT when only 1 photon/electron is seen, which is crutial to know when trying to calibrate the light that a PMT sees). To correct for this we need to only select the height point of any pmt pulse. This is done by calling the LandauFit code. This is a c++ code that first needs toi be compiled using `bash pipeline_LandauFit.sh`. The code needs to be recompiled every time it is changed for the changes to apply. No need to worry about the purple warnings. The fitting code is then called with `./LandauFit -i path_to_input_root_file -o path_to_desired_output_file.root`. The input file needs to be the one you have hadded previously. If you do not give any output, the corrected file will be saved as `Data/output_LandauFit.root`. Have a read through the comments in the `LandauFit.cc` code to understand what it is doing.

Have a look at the output of the landau fit code, you should have a histogram called PMT1_Normal, identical to the one you input and the corresponding corrected one. If you took dark rate data and have enough statistics you should be able to see the pedestal large value and a smaller peak further up in voltage, this is the 1 photo-electron peak.

3 Data collection with laser

3.1 Set up the laser/diffuser ball

3.2 Set-up trigger on Laser syncout

3.3 Wire multiple PMTs at once

3.4 Modify the code to account for multiple PMT used at once

4 Further data analysis

4.1 Bellamy fit to the PMT Voltage spectrum and gain calibration

4.2 Transient time spread measurements

4.3 Geometrical corrections

4.4 PMT efficiency correction